

Mend-Amar Badral

 MendeBadra |  Mend-Amar Badral |  mendebadra.github.io
 mendamar.badrall@gmail.com |  +36 70 7385 744 |  Budapest, Hungary

Driven by a problem solving mindset and fueled by curiosity, I aspire to develop genuine solutions for the real-world, impactful problems.

SKILLS

Languages: English (C1 [IELTS 8.0](#) )
Programming: Python, Julia, C/C++, R, SQL
Tools: Linux, Bash, Git, OpenCV, pandas, PyTorch, sklearn, open3d, YOLO, Docker, nerfstudio Quarto, $\text{\LaTeX}$

EDUCATION

Budapest University of Technology and Economics (BME) Sep, 2025 – June, 2027
M.Sc. in Computer Science Engineering
Budapest, Hungary
Supervisor: [Márton Vaitkus](#) 
Specialization: Data Science & Artificial Intelligence

National University of Mongolia (NUM) Sep, 2021 – June, 2025
B.Sc. in Applied Mathematics
Ulaanbaatar, Mongolia
GPA: 3.2/4.0 (**rank 3/34**)
Supervisor: [Galtbayar Artbazar](#) 
Thesis: [Evaluating land degradation using image processing methods](#) 

EXPERIENCE

Center of Mathematics Application, NUM Oct, 2023 – June, 2025
Undergraduate Research Intern

- **Linux:** Configured and maintained lab server remote access via SSH and Tailscale; managed access keys, automated system backups, and deployed containerized workflows in CLI-driven Linux environments.
- **C/C++ Development:** Built foundational programming knowledge through university coursework, implemented gradient descent algorithm in C/C++.
- **Scripting:** Developed [Python CLI tools](#)  for multispectral image processing using OpenCV; implemented multi-threading to cut processing times by 80% while managing version control in Linux. Obtained highly accurate vegetation index.
- **Infrastructure:** Replaced proprietary processing pipelines with open-source Docker workflows, tuning processing parameters and troubleshooting environment dependencies to ensure reliable execution.
- **Data Collection:** Participated in a field survey drone imagery captures on pastoral lands of different levels of land degradation.

PROJECTS

[aiSim LiDAR point cloud sensor fusion for 3D Gaussian Splatting](#)

- Developed a Python scripts to colorize LiDAR point cloud measurements obtained from aiMotive's [aiSim](#)  vehicle simulation platform to train 3D Gaussian Splatting in [nerfstudio](#) .

[Deep Learning Foundations and Concepts Book Examples Reproduction](#)

- Reproduced polynomial regression experiments from Bishop & Bishop's *Deep Learning Foundations and Concepts* using Julia notebooks to using interactive sliders.

[Exploratory Data Analysis – Bike Sharing Dataset](#)

- Performed exploratory data analysis on bike sharing dataset. Published the results on the personal blog using Quarto.

AWARDS

Stipendium Hungaricum Scholarship

Sep, 2025